A Review of User and Entity Behaviour Analytics for Insider Threat Detection: Detection Models, Evaluation Practice and the Security Operations Centre

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Abstract

User and Entity Behaviour Analytics (UEBA) looks for compromised accounts, malicious insiders and lateral movement by learning how users and devices normally behave and flagging departures from that baseline. Insiders use valid credentials and permitted actions, so signature-based tools see little of what they do. This paper reviews thirteen recent studies: nine that propose or evaluate detection and triage methods, and four that examine how analysts in Security Operations Centres (SOCs) handle the alerts such methods produce. After introducing the methods by technique (clustering, autoencoders, sequence models, graph models and contextual triage), the review compares them across four problems that cut through all of them: where the behavioural baseline comes from, what is profiled and at what granularity, how much labelled data is assumed, and how a score becomes an alert an analyst can act on. The comparison shows that headline results are hard to compare because data sets, label budgets and metrics differ from paper to paper; that the strongest benchmark figures depend on labels that deployments rarely have; and that a detector can rank events almost perfectly while producing an alert stream that is nearly all false alarms. The SOC studies locate the practical limit in alert volume, missing context and uninterpretable scores. Open problems include baseline poisoning, joint profiling of users and devices, and evaluation with working analysts.

Keywords: *user and entity behaviour analytics; insider threat detection; anomaly detection; autoencoder; graph neural network; security operations centre; false positives; explainability*

1. Introduction

Most data breaches involve a human element such as stolen credentials, misuse of privilege or error [1], and insider incidents are among the most expensive that organisations face [2]. They are also among the hardest to detect. An insider, or an attacker who has taken over an insider's account, logs in with valid credentials and opens files that account is allowed to open. No exploit runs and no signature matches, so detection based on known indicators of compromise has little to work with [3].

User and Entity Behaviour Analytics takes a different route. It learns a profile of normal behaviour for each user and each entity (a workstation, server or other device) from historical logs, and reports activity that departs from it [3, 4]. The attack does not need to be known in advance. Recent work has applied fuzzy clustering [4], deep autoencoders [5, 6], recurrent sequence models [7], graph convolutional networks [8, 9] and temporal graph models [10] to the problem, most often evaluated on the synthetic CERT insider threat data sets [11].

Detection is only half of the story. A separate line of research studies what happens once an alert reaches a SOC. Analysts report that the great majority of alarms are not actionable [12]. Alert quality is shaped by rule tuning long before any learning is applied [13]. Junior analysts often triage without a structured process [14], and many do not understand or trust the scores that machine-learning tools give them [15]. Systems that automate part of the triage have appeared in response [16]. Reviews of UEBA usually cover the first literature and ignore the second. We read them together, because a detector's value in practice is set by whether its alerts get acted on.

Section 2 gives background on UEBA, its data sets and its metrics. Section 3 introduces the nine detection and triage studies, grouped by technique. Sections 4 to 7 compare them across four problems: the source and stability of the baseline, what is profiled, the supervision assumed, and the path from score to alert. Section 8 brings in the four SOC studies and asks what they imply for the detectors. Section 9 tabulates the comparison, Section 10 lists open research gaps and Section 11 concludes.

2. Background

2.1 Insider threats and the UEBA approach

An insider threat is a risk posed by someone with legitimate access: an employee copying data before they leave, an administrator sabotaging systems, or an outside attacker using a stolen account. UEBA grew out of user behaviour analytics by adding entities such as hosts and network devices to the things being profiled [4]. It rests on one assumption: most users follow similar routines over long periods, so a baseline can be learned and current behaviour compared against it [3]. Typical signals are logon times and hosts, file and removable-media activity, e-mail, web browsing and authentication between machines.

2.2 Unsupervised anomaly detection

Labelled insider attacks are scarce, so most UEBA systems learn without labels or with very few. Isolation forests separate outliers using few random splits [17]. The local outlier factor compares the density around a point with that around its neighbours [18]. Autoencoders learn to compress and reproduce normal data [19], and inputs they reconstruct badly score as anomalous. LSTM networks [20] learn which event should come next; graph neural networks [21] learn from the links between users or hosts. All of these learn the baseline from data assumed to be mostly benign, so which data is used, and how old it is, matters a great deal.

2.3 Benchmark data sets

Real behaviour logs are personal data and are rarely released [3]. The CMU CERT insider threat test data sets [11] fill the gap with synthetic logs of a simulated organisation, covering logons, removable devices, files, e-mail and web traffic, with injected insider scenarios. Release r4.2 simulates 1,000 employees of whom 70 are malicious; r6.2 simulates 4,000 employees with five malicious insiders [8]. The Los Alamos (LANL) data set [22] is real: about 1.6 billion authentication, process and network events from 58 days of an enterprise network, with labelled red-team activity [10]. NSL-KDD [23] is still used, although it holds network connection records rather than user behaviour.

2.4 Metrics and the base-rate problem

Papers report accuracy, F1, true and false positive rates, the area under the ROC curve (AUC) or the equal error rate. With malicious activity a tiny fraction of the whole, these figures can mislead. Axelsson showed that when attacks are rare, even a very small false positive rate yields far more false alarms than true ones [24]. Accuracy is dominated by the benign class, and AUC measures ranking, not workload. Precision, or false alerts per day, answers the question an analyst actually has. Few papers report it.

3. Detection and Triage Approaches

This section introduces each method briefly. The critical comparison follows in Sections 4 to 7.

3.1 Similarity and clustering baselines

Landauer et al. [3] release CLUE-LDS, real logs from a cloud file-sharing platform with more than 5,000 users over more than five years, anonymised so that user, action, time and context survive. Their detector compares each user's daily event-type counts with learned profiles and retrains when behaviour changes.

They simulate account hijacking by swapping the identifiers of similar users. At the chosen setting it reaches a true positive rate of 65% at a false positive rate of 4.6%, under two false alarms a day.

Cui et al. [4] target a different weakness: clustering methods that place each sample in a single group and judge it only against that group. Their MAD-FPC model gives every entity fuzzy membership in several clusters. Behaviour is weighted by behaviour frequency-inverse entity frequency (BF-IEF), which works much like TF-IDF and boosts rare actions, and particle swarm optimisation searches for the centroids. An entity's anomaly factor is computed across all its clusters and thresholded with a box-plot rule. On NSL-KDD the method reaches an F1 of 0.93.

3.2 Reconstruction models

Fuentes et al. [5] apply deep autoencoders to the logs of a financial institution, embedding text fields such as process information with Doc2Vec [25] alongside numeric features. Training records are those the bank's filters had already cleared, and the authors treat them as possibly contaminated rather than clean. Per-feature reconstruction residuals show which fields drove each alert. Zhu et al. [6] take the autoencoder in another direction. AUTH builds separate models for time features and event-frequency features, each an adversarial autoencoder [26] with a temporal convolutional network and an LSTM in the encoder. Adversarial training pulls the latent space towards a normal distribution. On CERT it reaches an AUC of 0.932 with no labels at all.

The two papers share a model family but little else. Fuentes et al. accept a weaker evaluation (they had too few labelled anomalies to compute standard metrics) in exchange for real data and an explanation attached to every alert. Zhu et al. obtain a clean benchmark number on synthetic data, and their alerts come with no explanation at all.

3.3 Sequence models

Villarreal-Vasquez et al. [7] start from the observation that insiders hide attack steps among valid actions, and that detectors lose track of order beyond a certain sequence length. LADOHD trains LSTMs [20] on endpoint detection and response (EDR) events and judges each event against the sequence that preceded it. On 38.9 million events from 30 machines over 20 days, which include a four-day insider attack, it reaches a true positive rate of 97.29% at a false positive rate of 0.38%. LSTMs also ranked expected events in variable-length sequences better than hidden Markov models.

3.4 Graph models

Three studies use graphs, and each builds a different one. Li et al. [8] connect users. DD-GCN fuses original relationships with feature similarity, learns embeddings from both structures with attention, and reports accuracies of about 98.21% on CERT r4.2 and 98.65% on r6.2 with 60% of nodes labelled. Cai et al. [9] connect activities. LAN models the order of activities in each sequence and, by graph structure learning, links between activities in different sequences; a hybrid loss mixes self-supervision from normal activities with supervision from abnormal ones. It detects threats in real time and beats nine baselines by at least 9.92% (r4.2) and 6.35% (r5.2) in AUC. King and Huang [10] connect hosts. EULER treats lateral movement, a stage of the MITRE ATT&CK framework [27], as unlikely new edges in a sequence of authentication graphs. A graph network stacked on a recurrent encoder, distributed across machines, reaches an AUC of about 0.99 on LANL.

3.5 Contextual triage

Van Ede et al. [16] work downstream of detection, on the events existing detectors raise. DEEPCASE learns the context preceding each event with attention, clusters events with similar context, and asks analysts to label a few per cluster; the labels then spread to the rest. On 10.5 million events from 20 organisations

collected over five months, it filters 86.72% of events automatically and cuts manual workload by 90.53%, while underestimating risk in fewer than 0.001% of cases.

4. Where the Baseline Comes From

Every method in Section 3 learns normal behaviour from data, so its results are only as credible as that data. The reviewed studies fall into three groups, and each has a different blind spot.

DD-GCN, LAN and AUTH are evaluated on CERT [8, 9, 6]. CERT is public and labelled, which makes it the only setting where several methods can be compared at all. It is also synthetic. Landauer et al. [3] argue that simulated users are scripts that repeat similar actions, so false alarm rates measured on synthetic logs are likely to be too low. Their own real data shows a second problem that CERT does not capture: users' patterns drift over five years as systems and habits change. High accuracy on CERT therefore tells us that a method separates the injected scenarios from a regular simulated background. It says much less about how often the method would raise false alarms on a real, messier workforce.

Fuentes et al., Villarreal-Vasquez et al. and van Ede et al. use real data from a bank, a 30-machine commercial network and 20 organisations respectively [5, 7, 16]. Their findings are more convincing as evidence of real-world behaviour, but none of the data can be released, so no other method can be tested on it. LANL [22], used by EULER, is the exception: real and public, though limited to host-level events.

Cui et al. [4] sit apart. NSL-KDD holds network connection records labelled with attack classes, not user behaviour. A supervised random forest in their own comparison reaches an average F1 of 0.91, close to the 0.93 of MAD-FPC. The data set is easy to separate even without behavioural modelling. The F1 figure is therefore weak evidence about insider detection, even though the method's ideas, multi-cluster membership and rare-behaviour weighting, transfer well to UEBA.

How the baseline is kept current is a separate question. Landauer et al. retrain whenever behaviour changes, on 6.8% of days. The other detectors are trained once on a fixed split. Fuentes et al. are the only authors who say outright that their training data is probably contaminated. Everyone else implicitly assumes a clean baseline. This leaves a tension the literature has not resolved. Retraining handles drift, but it also gives a patient insider a way in: behave a little more unusually each week and the baseline will follow. Zhu et al. [6] list deliberate manipulation of the model as future work. None of the reviewed studies tests a slow-poisoning attack, and none tests a defence against one.

5. What Is Profiled, and at What Granularity

The 'E' in UEBA is honoured less often than the name suggests. DD-GCN and AUTH profile users [8, 6]. LADOHD profiles endpoint event streams [7], and EULER profiles hosts and the links between them [10]. Only MAD-FPC and the autoencoder of Fuentes et al. model users and entities together [4, 5]. Real insider cases cross that line all the time. Data theft involves a person, the machine they use and the removable device they plug in, so a detector that sees only one of these can miss the pattern that connects them.

The three graph studies show the same split from another angle. DD-GCN links users to users, LAN links activities to activities, and EULER links hosts to hosts. Each finds that relationships carry signal that isolated features miss. No study combines the three graphs, although an attacker moving from a compromised account to new hosts would leave traces in all of them. MAD-FPC's fuzzy membership [4] offers a cheaper form of the same idea, comparing an entity with several peer groups at once, without labels or a learned graph.

Granularity pulls in two directions. Cai et al. [9] argue that flagging a whole user or a whole day leaves analysts to search thousands of activities, and they move detection down to single activities in real time. LADOHD likewise classifies individual events. Finer decisions cut investigation effort per alert, but they

also multiply the number of decisions a detector makes, and with it the number of chances to be wrong. Section 7 shows where that leads.

6. How Much Supervision Is Assumed

The reviewed methods span a wide range of supervision. The detectors of Landauer et al., Cui et al., Fuentes et al., Zhu et al., Villarreal-Vasquez et al. and King and Huang use no attack labels for training [3, 4, 5, 6, 7, 10]. LAN adds a supervised term for labelled abnormal activities [9]. DD-GCN is semi-supervised, and its accuracy rises with the share of labelled nodes, reaching about 98% at 60% [8]. DEEPCASE needs analysts to label a sample from each cluster [16].

The headline numbers line up with this spectrum, which should make the reader cautious. The highest figures on CERT come from the methods that use the most labels. Labelling 60% of users is not a realistic budget for a security team that, by definition, does not know who its insiders are. The figures also resist direct comparison. DD-GCN reports accuracy on r4.2 and r6.2, LAN reports relative AUC gains on r4.2 and r5.2, and AUTH reports AUC and equal error rate on CERT. Different releases, metrics and label budgets mean that Table 1 cannot be read as a ranking. The CERT studies share no common protocol, and that gap is one of the findings of this review.

The way out may be the analyst rather than the data set. DEEPCASE shows that a few analyst decisions per cluster can be reused across millions of events [16]. Cai et al. reach the same idea from the other side, naming an interactive labelling framework as their next step because collecting abnormal labels is costly [9]. A detector that starts unsupervised and improves from the analyst decisions it already receives would fit how SOCs work. None of the detection papers evaluates one.

7. From Score to Alert

7.1 Ranking quality is not alert quality

EULER gives the clearest example in this review [10]. Its models reach an AUC of about 0.99 and detect 86-90% of malicious authentications at a false positive rate near 0.5%. Precision is only about 0.005 to 0.007. Legitimate authentications so outnumber the 518 malicious edges that more than 99% of alerts are false. This is the base-rate effect Axelsson described [24], and it applies to every method here. LADOHD's false positive rate of 0.38% [7] looks small. Yet an organisation logging a million events a day would face about 3,800 false events daily at that rate.

Landauer et al. are the only detection authors to report false alarms per day [3]. Their detector scores user-days rather than events, which keeps the count below two a day, but its true positive rate is only 65%. Read side by side, the two studies show the trade-off that Section 5 anticipated. Event-level detection finds more and pinpoints it better, at a cost in volume that AUC hides. Coarser units keep volume manageable but miss more. Reporting precision, or alerts per day, would let readers see where a method sits on this curve. Requiring several independent signals to agree before alerting is one way to move along it.

7.2 Triage after detection

DEEPCASE attacks the volume problem after the fact, removing most events from the analyst's queue [16]. It complements the detectors, since it takes existing alerts as input and needs no system-level information. Yet no reviewed study connects a UEBA detector to a triage stage like this and measures the result end to end. Grouping related alerts into a single case per user or incident, which DEEPCASE does by context, is also absent from every detector.

7.3 Explanation

Two of the nine methods explain their output. Fuentes et al. [5] report which features an autoencoder failed to reconstruct, an explanation built into the score. DEEPCASE [16] shows the events that formed the context of a decision. The sequence and graph models [7, 8, 9, 6, 10] return a score for a user, activity or edge, and nothing else. Graph models are a particular case. The structure that gives them their accuracy is also what makes their decisions hard to explain to an analyst.

8. The Analyst's View

The four SOC studies were not written about UEBA, but their findings bear directly on the detectors above.

Alahmadi et al. [12] surveyed 20 SOC practitioners and interviewed 21. Practitioners confirmed high false positive rates, manual validation and burnout. The authors found, though, that most so-called false positives are benign triggers. The alarm is technically correct, but the behaviour is legitimate in that organisation. For UEBA this is close to a definition of the problem: an anomaly detector flags unusual behaviour, and much unusual behaviour is legitimate. Precision measured against labels would count every such alert as a false positive, even when the detector did exactly what it was built to do. The authors' REACT model asks for alarms that are Reliable, Explainable, Analytical, Contextual and Transferable. By that standard, the seven of the nine reviewed methods that return a bare score fall short.

Vermeer et al. [13] interviewed 17 professionals about how network detection rules are written and managed. Rule specificity and the number of alerts and false positives a rule produces dominate those decisions. The authors criticise work that filters noisy alerts after the fact, arguing that it treats the symptom. This sits uneasily with DEEPCASE [16], which is exactly such a filter, and an effective one. The two positions can be reconciled. Vermeer et al. are right that a detector should control its own alert volume through thresholds, weights and suppression, and a triage stage can then deal with the remainder. For UEBA, the lesson is that volume control belongs inside the detector design, not only in a later stage.

Kersten et al. [14] gave ten tier-1 analysts a four-stage process for gathering evidence about an alert. Analysts who followed it were about 2.5 times more likely to reach a correct assessment, and the benefit was largest for alerts tied to real attacks. Oesch et al. [15] watched six US Navy analysts use two machine-learning tools on a cyber range. The analysts had no clear mental model of how scores were produced, mistrusted or misused them, and reached false conclusions at rates of 0.57 and 0.28 with the two tools. Experience and education did not predict performance. Between them, the two studies suggest that what accompanies an alert (its evidence, its context, a path for investigating it) affects decisions as much as the score does. None of the detection papers in this review evaluates its alerts with analysts, so this effect is unmeasured for every one of them.

9. Comparative Analysis

Table 1 summarises the detection and triage approaches. As Section 6 argues, the reported outcomes use different data, label budgets and metrics, and should not be read as a ranking. Table 2 summarises the SOC studies and their implications for UEBA design.

Table 1. Comparative summary of UEBA and insider threat detection approaches.

Approach	Data	Profiles / labels	Key reported outcome	Limitation noted
Adaptive similarity detector, CLUE-LDS [3]	Real cloud logs, 5,000+ users, 5+ years	Users; unsupervised	TPR 65% at FPR 4.6%; under 2 false alarms per day	Single source; simulated hijacking; no explanation

Approach	Data	Profiles / labels	Key reported outcome	Limitation noted
MAD-FPC fuzzy PSO clustering [4]	NSL-KDD	Users and entities; unsupervised	Accuracy 0.92, F1 0.93 (random forest 0.91)	Network data, not behaviour logs; static; unexplained
Deep autoencoder with Doc2Vec [5]	Real bank logs, numeric and text	Users and entities; unsupervised	Detects real and synthetic anomalies; per-feature explanations	Too few labels for standard metrics; private data
AUTH adversarial autoencoder [6]	CERT	Users; unsupervised	AUC 0.932; EER 0.146	No explanation or ranking
LADOHD LSTM [7]	EDR, 38.9 M events, 30 hosts	Endpoints; unsupervised	TPR 97.29% at FPR 0.38%	One organisation, one source; unexplained
DD-GCN [8]	CERT r4.2, r6.2	Users; semi-supervised	Accuracy 98.21% / 98.65%; F1 about 93%	Needs 60% labelled nodes; no explanation
LAN graph structure learning [9]	CERT r4.2, r5.2	Activities; hybrid supervision	AUC at least 9.92% / 6.35% above 9 baselines (real time)	Needs abnormal labels; no explanation
EULER temporal graph [10]	LANL, 1.6 B events	Hosts; unsupervised	AUC about 0.99; TPR 86-90% at FPR about 0.5%	Precision about 0.006; no explanation
DEEPCASE [16]	10.5 M events, 20 organisations	Alerts; semi-supervised	Workload reduced by 90.53%; under 0.001% risk underestimated	Needs analyst labels; works on existing alerts

Table 2. Summary of SOC practice studies and their implications for UEBA design.

Study	Method	Principal finding	Implication for UEBA
[12]	Survey (n = 20) and interviews (n = 21)	Most false positives are benign triggers; alarms lack reliability, explanation and context	Report benign triggers separately; attach explanation and context to every alert
[13]	Interviews with 17 professionals	Rule specificity and alert volume drive rule management; downstream filtering treats the symptom	Control alert volume inside the detector, not only by later filtering
[14]	Experiment with 10 tier-1 analysts	A structured four-stage process makes accurate assessment about 2.5 times more likely	Present alerts with structured evidence and a clear investigation path
[15]	In-situ study, 6 analysts, 2 tools	Analysts lack a mental model of ML scores; false conclusion rates 0.28-0.57	Scores must be interpretable; design and test with analysts

10. Challenges and Research Gaps

The comparison points to the following open problems.

- **A shared evaluation protocol:** CERT studies use different releases, label budgets and metrics [8, 9, 6], and studies on real data cannot be reproduced [5, 7]. A common protocol on public data, ideally including real logs such as CLUE-LDS [3], would make results comparable.
- **Operational metrics:** Precision and alerts per day should be reported next to AUC, since near-perfect AUC can coincide with precision below 1% [10, 24]. Benign triggers should be counted separately from false positives [12].

- **Realistic label budgets:** The strongest results use labelled users or activities [8, 9]. Methods that begin unsupervised and learn from a small stream of analyst decisions [16] are missing among the detectors.
- **Joint profiling:** Few detectors model users and devices together [4, 5], and no study combines user, activity and host graphs [8, 9, 10].
- **Baseline poisoning:** Baselines must adapt to drift [3], but adaptation opens a path for slow manipulation that no reviewed study tests [6].
- **Volume control and corroboration:** Detectors output a single score. Requiring agreement between independent signals before alerting, ranking alerts by risk and grouping them into cases [16] are rarely studied, and never together.
- **Explanation:** Seven of nine methods return an unexplained score, against clear evidence that analysts need explanation and context [12, 14, 15].
- **Evaluation with analysts:** No detection paper tests its alerts with working analysts, and no SOC study evaluates a UEBA detector [12, 13, 14, 15].

11. Conclusion

UEBA research has moved from clustering [4] and adaptive similarity [3] to autoencoders [5, 6], LSTMs [7] and graph models [8, 9, 10], and benchmark accuracy is now high. The comparison in this review qualifies that progress. Benchmark results come mostly from synthetic data, under label budgets that real teams do not have, and are reported with metrics that cannot be compared across papers. A detector can rank events almost perfectly and still send analysts an alert stream that is more than 99% false [10]. The useful ingredients exist separately: adaptive baselines [3], joint profiling of users and entities [4], built-in explanation [5] and analyst-driven triage [16]. No reviewed system brings them together.

The SOC studies explain why that matters. Analysts are limited by alert volume, missing context and scores they cannot interpret [12, 13, 14, 15], not by the last point of AUC. We suggest that UEBA work should report precision and alerts per day; evaluate on real or realistic data under a shared protocol; profile users and devices together; protect adaptive baselines against slow poisoning; control alert volume inside the detector; and explain each alert in terms an analyst can check. The largest gap is the simplest to state: detectors are not yet tested with the people who have to act on their alerts.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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